



A comprehensive survey on data aggregation techniques in UAV-enabled Internet of things

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ABSTRACT

In recent years, unmanned aerial vehicles (UAVs) have been used to extend the Internet of things (IoT) framework owing to their vast applications, monitoring and surveillance capability, ubiquity, and mobility. To support IoT requirements, UAVs must be capable of aggregating, processing, and transmitting data in real-time basis. As not only the number of IoT devices but also the amount of data to be collected is increased, data aggregation is of great importance. Recently, the UAV can also function as a mobile edge computing server in association with aerial data aggregation. This paper is the first to survey the various aspects and techniques of UAV-based aerial data aggregation for IoT networks. After addressing key design issues, we review the existing data aggregation techniques along with possible future direction. They are then compared with each other in terms of major operational features, performance characteristics, advantages, and limitations. Open issues and research challenges are also discussed with possible solution approaches.

1. Introduction

Recent technological advancements have led to unprecedented changes in the support roles played by unmanned aerial vehicles (UAVs) or drones, which were originally applied to military [1] and surveillance [2,3] operations. Their low cost and ubiquity [4], coupled with their basic working principles [5] of fast deployment, high mobility [6], ease of use, line-of-sight (LoS) connectivity [7,8] and reliable data communications [9], have led to their advanced support for civilian communications [10,11]. Applications now include wildfire monitoring [10], consumer payload delivery [12–14] image recognition [15], aerial base-station (BS) relay [16], search and rescue [17], and smart agriculture [18]. By acting as bridges between cyber and physical layers, they have unlocked seemingly limitless possibilities for autonomous operations [19]. Hence, it is no surprise that they are now being considered for extending Internet of thing (IoT) networks [20], which will further facilitate our society's low data footprint and federated processing needs [21,22].

According to a report published by Statista.com [23], the number of connected IoT devices is projected to reach more than 29 billion by 2030, three times more than the number of connections in 2020. Moreover, with recent developments in communication technology in

association with 5 G and beyond, it has been made possible to integrate fast moving UAV-based services into the existing communication system [24]. This increase is naturally expected to generate far more traffic than current networks can handle. Augmented by extant BSs, UAVs can assume the roles of computing centers or edge servers to process sensed data from ground IoT devices while reducing delays [25–28].

1.1. Motivation

Notably, most IoT devices are equipped with limited-capacity power supplies while being expected to function wirelessly for long periods [29,30]. Moreover, they are frequently installed in inaccessible or hard-to-reach locations, such as alongside the actuators of autonomous control devices. Hence, they are difficult to maintain, repair, and replace. Despite many challenges, UAVs demonstrate lots of promises in enabling many IoT services such as data aggregation [31].

Data aggregation, the process of compiling and structuring data for high-speed network transmission and software-level consumption [32, 33], decreases network overhead, improves reliability, reduces transmission delays [34], facilitates robust network behaviors, reduces redundancy, lowers energy consumption [35], and improves quality of service (QoS). Certain types of aggregation are also geared toward

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improving data privacy and security [36,37]. Particularly, the urgency of the need for continued advancement in these areas has been multiplied by the recent pandemic [38].

The energy consumed by IoT devices for data processing can be classified into three functional types: sensing, computing, and communicating. Generally, communication costs are much higher than combined sensing and computing costs [39]. Because aggregation improves the speed of processing large amounts of data, the overall energy consumption for the network can be reduced [40,41]. Therefore, researchers have sought to optimize these functions [42]; however, additional issues of data reliability and transmission delays have been encountered [34].

Traditional aggregation strategies are suitable for conventional fixed network infrastructures. However, some IoT devices are now mobilized (e.g., the internet of vehicles (IoV) [43]), which has resulted in scenarios in which network congestion and topologies have become largely unpredictable [44]. It was recently shown that UAVs can be employed as data aggregators [45] for both IoV and stationary IoT systems (even underwater). More importantly, during emergency scenarios, UAVs can be deployed quickly to restore data communications and provide on-demand coverage [46–48].

1.2. Related surveys

Several surveys have been conducted on data aggregation in IoT and wireless sensor network (WSN) environments. Table 1 provides a summary of these surveys. However, it should be noticed that none of them addressed aerial data aggregation or UAV-enabled IoT data aggregation explicitly. The bottom row of Table 1 summarizes our own survey for comparative purposes.

Table 1
Existing surveys on data aggregation.

Article Ref.	Year	Main theme			Key contributions	Limitations
		DA	UAV	IoT		
[49]	2017	●	–	●	Provides a systematic review of existing data aggregation techniques for IoT from the aspects of centralized, tree and cluster-based methods.	Aggregation mechanism of dynamic IoT nodes not discussed.
[50]	2019	●	○	●	Identifies and explains different data aggregation schemes in massive machine type communication (mMTC) access in 5 G cellular communication.	No conclusive remarks regarding aerial data aggregation.
[51]	2019	●	–	–	Studies data aggregation in underwater wireless sensor network (UWSN) based on cluster, non-cluster and other techniques which are further classified into sub-categories.	Neither IoT, nor aerial aggregation mechanism studied.
[52]	2020	●	–	–	Covers different data aggregation methods in wireless sensor network (WSN) and presents relative comparison between different aggregation methods.	Neither IoT, nor aerial aggregation mechanism studied.
[53]	2020	●	–	○	Infers the security aspects of data communication with and without data aggregation in WSN. Additionally, application scenario and security goals are categorized accordingly on secure data aggregation (SDA).	The concept of aerial data aggregation not discussed. Furthermore, aggregation of IoT device is lightly discussed.
[54]	2020	●	–	●	Discusses data aggregation namely in four different scenarios with their respective classification.	Unable to recognize UAV as potential aerial data aggregator.
[55]	2021	●	–	●	Reviews data aggregation in terms of security and data privacy and transmission mechanism for health monitoring fog-assisted IoT devices.	Potentiality of aerial data aggregator in healthcare is ignored.
[56]	2021	●	–	●	Presents a systematic literature review for data aggregation scheme based on mobile agent-based aggregation and client-server-based aggregation and makes a comparative study with each scheme.	UAV enabled data aggregation was not studied
[57]	2022	●	○	–	Summarizes the established compressive data aggregation techniques in terms of routing protocol, clustering methods and combination of various techniques.	No deeper studies regarding aerial data aggregation.
[58]	2022	○	●	●	Provides a review of the scenarios and key technologies involved in UAV-assisted data collection for IoT in terms of sensor clustering, UAV data collection modes, joint path planning, and resource allocation.	No in-depth analysis of UAV-assisted IoT data aggregation scenarios is studied.
[59]	2023	○	●	●	Studies UAV-assisted data collection schemes, design issues, implementation challenges, and application scenarios for data collection using UAVs as an auxiliary device.	No specific data aggregation scheme using UAVs for IoT devices got the limelight.
This Survey	2023	●	●	●	Elicits the key aspects of aerial data aggregation in terms of UAV role management, UAV-IoT communication scheme, different design issues, existing works, and future recommendations and prospects in the area of interest.	

Note: ● Studied; ○ Partially studied; – Not studied.

The authors of [49] provided a systematic review of the advantages and disadvantages of existing data aggregation techniques for IoT from the perspectives of centralized, tree, and cluster networks. A variety of performance metrics were applied, but the review in question did not address dynamically changing IoT nodes.

The survey conducted in [50] identified data aggregation schemes for massive communication access in 5 G cellular networks. They mentioned the prospects of UAVs as data aggregators but were unable to further examine the possibility. The survey from [51] provided a comprehensive study of underwater sensor network (UWSN) data transmissions and processing techniques by comparing solutions with and without aggregation. They also examined factors impacting aggregation, such as packet size, delay, delivery ratio, packet drop, and energy consumption. The review presented in [52] compared several WSN aggregation methods, classifying them into centralized, distributed, and hybrid types and suggesting algorithmic improvements and evaluation matrices for each. The study in [53] discussed the security aspects of data aggregation and provided a comparison of methods and topologies.

According to [54], there are four data aggregation media techniques: terrestrial WSNs, UWSNs, wireless underground sensor networks, and wireless body sensor networks. The researchers studied the challenges and advantages of data aggregation in network architectures, but they did not consider aerial data. The authors in [55] analyzed a secure fog-assisted aggregation mechanism for healthcare IoT in which the strengths and weaknesses of compressed data collection were studied alongside secure data aggregation techniques. In [56], the authors reviewed extant IoT data aggregation techniques and discussed cases from client–server and mobile agent perspectives. However, they did not address UAV-enabled networks. Compressive or compressed sensing (CS) for data aggregation purposes was studied in [57], and three

aspects (i.e., routing, clustering, and combined) were identified, resulting in UAV-aided aggregation schemes based on reinforced learning (RL) and genetic algorithms (GA).

The survey work in [58] focuses on different data collection schemes for IoT devices using UAVs and discusses the different key aspects such as arrangements of sensors, different modes for data collection, a combined analysis of resource management, and trajectory of UAVs. The recent survey in [59] covers the generalized aspects of data collection schemes in a detailed manner, in which various data collection schemes are studied with respective advantages and limitations. In addition, the lessons acquired from each scheme are addressed and many of research challenges are introduced.

The interaction between UAVs and IoT devices for data-driven applications such as data collection and data aggregation follows very close relationship in terms of designing system models. Although many of operating procedures are similar in some respects for UAV-aided data collection and data aggregation, our survey mainly focuses on the UAV-enabled data aggregation techniques for IoT. The contribution of our survey is discussed in the following subsection.

1.3. Contributions

The purpose of our present survey is to examine the advantages, limitations, and lessons learned from current state-of-the-art UAV-based IoT data aggregation techniques explicitly and summarize our future research recommendations. We also propose a novel taxonomy of data aggregation techniques in UAV-aided IoT networks. The main contributions of this study are briefly described as follows:

- Comprehensive background information on UAV network communications that considers IoT data aggregation is summarized.
- Key design issues that account for data aggregation in underlying UAV-enabled IoT are addressed.
- Existing aerial data aggregation techniques are categorized based on the roles of UAVs and reviewed with their respective strengths and weaknesses. Some of the possible research directions are also discussed.
- The aggregation techniques are qualitatively compared in detail in terms of operation features, performance parameters, and design objectives.
- Open issues and research challenges are discussed along with possible future directions for future enthusiastic researchers.

To the best of our knowledge, this is the first survey to address UAV-enabled IoT data aggregation.

1.4. Organization of the study

The rest of this paper is organized as follows. In Section 2, background information regarding UAV-enabled data aggregation techniques is explained. In Section 3, we discuss key design issues of data aggregation in UAV-enabled IoT. In Section 4, data aggregation schemes are categorized according to their roles and subsequent algorithms. In Section 5, the data aggregation techniques are reviewed and discussed extensively. In Section 6, data aggregation techniques are qualitatively compared. In Section 7, open issues and research challenges are discussed along with future directions. Finally, concluding remarks are presented in Section 8.

2. Preliminary background

In this section, the preliminaries of UAV networks and aggregation techniques that can be combined for IoT environments are summarized.

2.1. Unmanned aerial vehicles

UAVs are autonomous, fixed-wing, or rotary remote-operated aerial vehicles. The scope of this review includes low-altitude (200–500 m) UAVs used for air-to-ground communications [7,19,60–64] that are applicable to both civilian and military operations. A brief comparison of different UAV types is presented in Table 2. Selection of the type of UAVs is very important depending on the mission requirements and application scenarios, which can have a significant impact on the system design aspects.

To understand UAV-assisted IoT data aggregation and design an effective aggregation scheme, we must first gain a clear understanding of single- and multi-UAV deployment scenarios. In some cases, mission objectives can be achieved by deploying a single UAV rather than deploying multiple UAVs and vice versa. Single UAV communicates with base stations (BSs), high-altitude platforms, and sensor nodes whereas, in multi-UAV networks, multiple UAVs work together in a centralized, distributed, or combined manner via shared connections to perform the same tasks. Table 3 lists the distinctions which are crucial for designing the system model of an aggregation scheme or mission budget.

2.2. Internet of things

Heterogeneous IoT environments require interconnectors [29,65] whose basic architectures are classified into four layers [65] (see Fig. 1): sensing, networking, cloud computing, and application. IoT are different from a WSN in terms of components and operations. The similarities and differences are studied in [59]. A brief discussion of each layer is as follows.

2.2.1. Sensing layer

This layer comprises the sensors used to accept and process data, preparing them for transmission over the network. Conventional sensor types usually comprise acquisition, sink, and management nodes. BS are an exemplary type of sink.

2.2.2. Networking layer

This layer accounts for the movement of data between nodes (e.g., single- or multi-hop topologies). Dynamic network topology reorganization is crucial for IoT congestion reduction [66]. Energy-saving protocols are essential to IoT lifecycle support [67].

2.2.3. Cloud-computing layer

This layer provides heavy analytical processes that would otherwise require too many resources from basic IoT devices and nodes (edges) [68]. Cloud computing has been demonstrated to mitigate traditional middleware computing techniques that rely on proprietary protocols and onboard computing, which are prone to delays and memory overhead.

Table 2
Comparison of fixed and rotary wing UAVs.

Feature	Fixed-Wing	Rotary-Wing
Hovering over fixed position	No	Yes
Flexibility in trajectory design	No	Yes
Energy-Efficiency in trajectory planning	Higher	Lower
Payload capacity	Higher	Lower
Movement speed	Higher	Lower
Flight time	Longer	Shorter
Advantageous application case	Long-term monitoring or surveillance	On-demand emergency scenario

2.2.4. Application layer

This layer is responsible for the user interface and data visualization [69]. The goal is to provide an easy and accessible interface with which operators can utilize the maximum potential of the IoT in everyday life.

2.3. UAV-aided data aggregation

When UAVs are deployed as data aggregators, they often act as the middle layer (relaying or routing) between IoT devices and cellular BSs. In such scenarios, the UAV performs essential network services so that nodes and edges can expend minimal energy. Unlike IoT devices themselves, UAVs can be recharged or refueled as often as necessary to continue the mission.

There are three modes of UAV data aggregation: cruise (token node availability), loiter (persisting above a certain node), and dedicated (specific nodes along a flight path). In this case, cluster heads (CHs) are treated as aggregation locations. Detailed scenarios are discussed in subsequent sections.

3. Key design issues of data aggregation for UAV-enabled IoT

In this section, we address key design issues of data aggregation in UAV-enabled IoT. Researchers have worked for years to mitigate performance bottlenecks while improving data aggregation effects. However, no single technique provides all desired benefits at once; hence, trade-offs are a foregone conclusion.

3.1. Number and type of deployed UAVs

Efficient UAV data aggregation, in the simplest terms, requires appropriate numbers and types of UAVs based on the mission requirements and target environment (e.g., performance parameters, data types, areas of interest, and node coverage). The control system can be either centralized or distributed. Centralized systems govern all the management and processing routines of all devices, including the UAV. On the other hand, decentralized (distributed) systems use autonomous and cooperative components [70]. Both types have strengths and weaknesses. Notably, a centralized system has inherent single points of failure, and distributed systems entail distributed bottlenecks.

3.2. Communication schedule

A network communication schedule refers to the planned route of the UAV to the proximity of IoT devices in a certain order. Collection and retransmission requirements must be well accounted for in the flight plan, as UAVs have limited memory, fuel, and computing capacities; hence, network load balancing often relies on complex physical aerospace logistics. This makes it difficult to react to emergency priorities

Table 3
Comparison of single- and multi-UAV networks.

Parameter	Single-UAV Networks	Multi-UAV Networks
Coverage Area	Smaller	Larger
Mission Completion time	Longer	Shorter
Chances of mission completion in case of failure	Poor	High, can be reconfigured
Network System	Centralized	Centralized, distributed or hybrid
Relative cost	Higher	Lower
Control Complexity	Lower	Higher
Bandwidth requirement	High	Medium
Security	Low	Medium
Antenna type	Omni-directional	Directional
Scalability	Limited	High

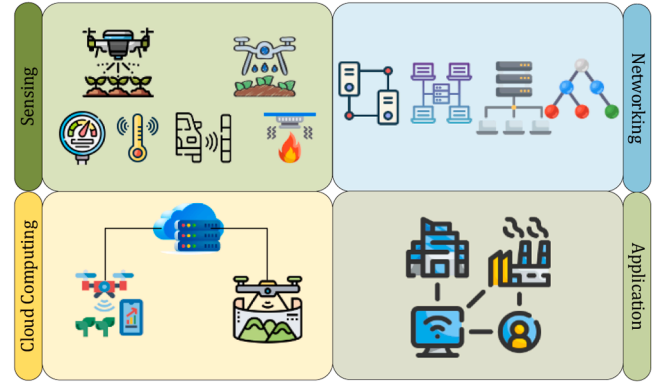


Fig. 1. Basic IoT-enabled system architecture.

because the dynamic rescheduling of such networks quickly requires nonlinear computations.

3.3. Device activation period

IoT device activation relies on traffic control, weather sensing, and fire detection control factors, among others. Some IoT sensors are perpetually reactive (e.g., fire and smoke detectors), whereas some are periodic (e.g., weather monitoring and smart metering). The UAV positioning scheme must therefore be designed to match these IoT device activation schedules, given that “random” activations cannot be scheduled [71]. The Third Generation Partnership Project (3GPP) defines the applicable temporal factor, $t \in [0, T]$, as follows :

$$D(t) = \frac{t^{x-1}(T-t)^{y-1}}{T^{x+y-1} B(x, y)}, \quad (1)$$

where $[0, T]$ is the time interval in which an IoT device is active, and $B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt$ is the beta function, whose parameters are denoted as x and y [72].

Most periodic activation schemes are deterministic, meaning that activation periods are known prior to network design. For these, device activation is denoted as $\left[\frac{T}{\tau_i}\right]$ over the period $[0, T]$ [72], assuming that IoT device i remains active for every τ_i seconds. In many schemes, the device activation model is considered as an assumption for the simplicity of problem formulation.

3.4. Travel time of UAVs

Like other UAV-supported systems, data aggregation is subject to limited travel times and sustainment power. A typical rotary-wing UAV has a flight time of ~ 40 min [73]. Hence, the urgency of IoT data communication factors heavily into the given window of opportunity. IoT sensors often begin transmitting data when the UAV arrives in proximity; hence, the flight plan must account for their transmission ranges and their necessary order of operation [74,75]. Thus, the total travel time is expressed as

$$T_{tot} = \sum_{k=1}^N T_{hov}^q + T_{tra}, \quad (2)$$

where T_{hov}^q is the loitering time at the designated hovering location, q , and T_{tra} is the travel time, which includes the coverage time and trajectory in the area of interest.

3.5. Cruise control and trajectory

UAVs are highly agile; thus, when using them as data mules, the efficiency of waypoint planning is the most essential design consideration, and static or dynamic trajectories can be determined [76]. Solving the waypoint scenario is often formulated as the traveling salesman problem (TSP) or the vehicular routing problem (VRP) [77–79]. In the TSP, UAVs visit all the required nodes using the shortest possible path, followed by returning to the origin. Notably, TSP is an NP-hard problem, and its complexity increases with the number of waypoints. Various techniques have been employed to solve the TSP for UAV flight operations [80], including deep reinforcement learning (DRL) methods [81]. Trajectory design can be either preplanned or unplanned. Preplanned trajectories are trained offline and/or controlled from the central base station, whereas unplanned trajectories are trained online to handle dynamic scenarios.

3.6. Communication mode

There are many ways in which UAVs can establish communications with IoT devices (e.g., Wi-Fi, Long-Term Evolution (LTE), Bluetooth, and ZigBee). Both Wi-Fi (IEEE 802.11 g) and ZigBee (IEEE 802.15.4) functions in a 2.4-GHz frequency bandwidth, causing both to suffer interference from similar devices. Because UAV-aided IoT environments often deal with bursty traffic [71], a larger Wi-Fi overhead can lead to improper spectrum utilization. ZigBee devices using direct sequence spread spectrum techniques can mitigate this issue. However, ZigBee struggles to adapt to fast-changing mobile communication environments. LTE can deal with nearly all of the above issues; however, in the case of disasters and other emergencies, existing infrastructures may cease to exist. A generalized overview of the different communication protocols with respective data rate and communication range is discussed in [82].

Bluetooth 5.0, more commonly known as Bluetooth Low Energy (BLE), supports long-distance communications and UAV-aided IoT environments [83]. UAV-assisted BLE can transmit data at faster rates while providing a longer range of coverage than ZigBee or Wi-Fi.

The Low Range (LoRa) physical proprietary communication technique is employed in sensory networks for long-range communications. A UAV swarm can use LoRa to provide services for aggregation-related tasks [84]. 3GPP has imposed guidelines for these kinds of UAV-aided sensory data processing and service tasks [85].

3.7. Energy

The energy expended by IoT node a can be expressed as

$$E_a^{Tot} = E_a^s + E_a^p + E_a^t + E_a^r, \quad (3)$$

where E_a^s is the energy required for sensing data, E_a^p is the energy required for processing and aggregating data, E_a^t is the transmitting energy, and E_a^r is the expended energy for receiving data. For UAV data aggregation, the energy of the UAV is expended in three major areas: loitering, aggregation-related tasks, and communicating with peripheral devices [86]. Moreover, the energy required for communication relies on the transmission data size, distance between source and destination, and the number of UAVs and IoT devices in the network. Researchers have mainly focused on mitigating only one or two of these issues at the same time.

3.8. Channel model (ground-to-air)

For any UAV-supported communication model, it is necessary to account for stochastic transmission behaviors occurring with LoS and non-LoS (NLoS) services. Several channel models were examined in [87], in which the UAV maintained LoS communications between two or more distributed devices that were otherwise blocked from one another.

The generalized expression of LoS probability between ground IoT devices a and b [72] is defined as follows:

$$P_{Los}^{ab} = \frac{1}{1 + \alpha \exp(-\beta[\theta_{ab} - \alpha])}, \quad (4)$$

where α and β are constants that depend on environmental factors and carrier frequency, respectively. θ_{ab} is the angle of elevation between the ground device, a , and the UAV, b , which is calculated as

$$\theta_{ab} = \frac{180}{\pi} \times \sin^{-1} \left(\frac{h_b}{d_{ab}} \right), \quad (5)$$

where h_b is the altitude of the UAV, and $d_{ab} = \sqrt{(x_a^{IoT} - x_b^{UAV})^2 + (y_a^{IoT} - y_b^{UAV})^2 + h_b^2}$, which is the distance between IoT device a and UAV b [72]. Thus, increasing the UAV altitude or angle of elevation increases the LoS probability. The NLoS probability can be defined as $P_{NLoS}^{ab} = 1 - P_{Los}^{ab}$. The path-loss model for both LoS and NLoS communications between IoT device a and UAV b is expressed as follows:

$$L_{ab} = \begin{cases} \eta_1 \left(\frac{4\pi f_c d_{ab}}{c} \right)^\gamma, & \text{LoS link} \\ \eta_2 \left(\frac{4\pi f_c d_{ab}}{c} \right)^\gamma, & \text{NLoS link} \end{cases}, \quad (6)$$

where γ is the path-loss component, f_c is the carrier frequency, and η_2 and η_1 are the excessive path-loss components of the NLoS and LoS links, respectively [72]. As in [69], the transmission rate between IoT node a and UAV b is expressed as

$$R_{tx}(a, b) = B \log_2 \left(1 + \frac{|h(a, b)|^2 P_a}{N_0} \right), \quad (7)$$

where B is the bandwidth of the channel, $h(a, b)$ is the channel gain between IoT node a and UAV b , P_a is the transmission power of IoT node a during communication, and N_0 is the additive white Gaussian noise.

3.9. UAV power model

The UAV expends energy when moving vertically and horizontally, given four corresponding forces: lift, drag, thrust, and weight [88]. For a multicopter UAV hovering or moving vertically, the power consumption is

$$P = (M + m)^{3/2} \sqrt{\frac{g^3}{2\sigma A N}}, \quad (8)$$

where M is the weight of the UAV frame, m is the combined weight of the payload and battery, g is the gravitational acceleration, A is the area of the rotor disk, and σ is the air fluid density [89]. The energy expended by a typical fixed-wing UAV moving horizontally is formulated as

$$P = \frac{2\kappa L^2}{\sigma \pi V b^2} + \frac{C_D S \sigma V^3}{2}, \quad (9)$$

where L is the lift force, C_D is the drag coefficient, S is the wing area, σ is the air fluid density, V is the velocity of the UAV, and b is the wingspan [90].

In some of the recent works, the propulsion power of a rotary-wing UAV with respect to velocity v is defined as follows [91]:

$$P(v) = P_1 \left(1 + \frac{3v^2}{U_{tip}^2} \right) + P_2 \left(\sqrt{1 + \frac{v^4}{4v_0^4}} - \frac{v^2}{2v_0^2} \right)^{1/2} + \frac{1}{2} d_0 \rho g A v^3 \quad (10)$$

The above equation consists of three parts: blade profile, induced power, and parasitic power. P_1 and P_2 are the blade profile power and

induced power, respectively, under hovering condition (when $v = 0$). U_{tip} is the speed of the rotor blade tip and v_0 is the mean rotor induced velocity while hovering. d_0 , ρ , g , and A are denoted as fuselage drag ratio, density of air, solidity of rotor, and disk area, respectively. The energy consumption of UAV over the interval of T can be expressed by

$$E_{UAV} = \int_0^T P(v) dt. \quad (10)$$

3.10. Trajectory constraint of UAVs

In order to ensure that UAV begins its journey from the central station, aggregates data by visiting the waypoints, and finally returns back to the central station, several constraints are imposed which are expressed as

$$\sum_{i=1}^K x_{i,j}^U = 1 \quad i \neq j \quad \forall j \in N \quad (11)$$

$$\sum_{j=1}^K x_{i,j}^U = 1 \quad i \neq j \quad \forall i \in N \quad (12)$$

$$\sum_{j=1}^K x_{0,j}^U = \sum_{j=1}^K x_{i,K+1}^U \quad 1 \quad i \neq j \quad \forall i, j \in N \quad (13)$$

where $x_{i,j}^U$ is a binary variable, $x_{i,j}^U \in \{0, 1\}$ indicates the condition of UAV traverse from one hovering point to another. $x_{0,j}^U$ represents the initial starting constraint for the UAV, and $x_{i,K+1}^U$ represents the condition of return to the desired location.

3.11. Routing and medium access control (MAC) protocols

Routing and MAC protocols are crucial to determining the best flight path for effective aerial data aggregation. The routing scheme facilitates network scalability and energy conservation under two basic UAV-aided IoT scenarios: among UAV and IoT devices. For multi-UAV scenarios, UAVs require a proper routing scheme to maintain communications among themselves [92], often entail centralized or distributed methods [93].

Several routing schemes and MAC protocols have been examined for these purposes [6,11,13,62,63,94]. Recently, the development of a UAV-based energy-efficient (UEE) routing protocol for low-power and lossy networks (RPL) was proposed to meet the low energy requirements of UAV-aided IoT communications [95].

4. Taxonomy of data aggregation techniques

In this section, a taxonomy of data aggregation techniques in UAV-aided IoT networks is established. Our taxonomy is based on the roles of UAVs as shown in Fig. 2. Determining the roles of UAVs in aerial data aggregation is a critical design consideration. A visual demonstration of the different UAV-aided data aggregation techniques is illustrated in Fig. 3, where UAVs perform the integrated roles for aerial data aggregation. Each of the roles of UAVs is discussed in the following four subsections.

4.1. Direct aggregation

A UAV can perform aggregation tasks by itself, using its onboard processing capability. A typical scenario would involve UAV loitering over a collection of designated ground nodes in the area of interest. Trajectories and waypoints can be applied based on the arrangement of nodes, their computational power, etc. After aggregation, the UAV sends the aggregated data to a ground device for further processing. In most cases, a single UAV is deployed at a time.

There are several advantages to this scenario. For example, if the UAV can perform aggregation, the ground IoT devices will require less processing and fewer computations, which extends their lifetimes.

However, UAVs must be loaded with greater storage and computational power, causing them to require more visits to charging stations.

4.2. Aggregation as collector

UAVs can collect aggregated or raw sensing data from distributed ground IoT devices and CHs in the field of interest and relay them to sinks or BSs. Flight plan and trajectory considerations are roughly the same as before, as are the advantages for ground nodes, in that they are not burdened with additional processing requirements. The UAV, on the other hand, does not require extra computational and processing power as it is solely responsible for data collection and relay.

4.3. Aggregation as sink

UAVs can be deployed as aerial sinks or BSs, in which they perform the roles of aggregation and extended processing. To fulfill this role, UAVs must have very strong onboard computational and processing power. This is very beneficial to regions of interest lacking communication infrastructures (e.g., post-disaster recovery). In such cases, the need for multiple simultaneous UAVs would be greater, further increasing the logistical complexity.

4.4. Hybrid aggregation

UAVs will likely perform a combination of the roles described above, even simultaneously. Naturally, the computational power and processing capability would need to be very robust, as would the flight planning capacity of the command-and-control infrastructure. Fig. 3 presents a role-based aggregation scheme in which different UAV types are deployed in rural and urban areas to perform related aggregated tasks.

5. Data aggregation techniques

In this section, existing UAV-enabled data aggregation techniques are extensively reviewed. The various UAV roles in IoT data aggregation are exemplified below, using reports from the literature. Then, the discussed techniques' respective strengths and weaknesses are discussed along with lessons learned.

5.1. Machine learning (ML)-Based data aggregation

ML approaches to data aggregation and processing are quite popular among disparate research fields. However, for aerial data aggregation, there is currently a lack of sufficient data that can be used for model training. To address this issue, DRL is a viable option for handling unstructured or unknown features in which knowledge discovery is expected during operations. Unlike supervised learning, in which a labeled dataset is trained for model prediction, and unsupervised learning, in which unlabeled data are analyzed on the fly, DRL interacts with unknown datasets from scratch as they explore the region of interest, working toward a desired algorithmic reward by applying a trial-and-error method. For UAV flight planning, DRL can be used to design optimal trajectories and data-aggregation procedures.

Federated learning (FL) is another approach as it benefits from decentralized training data consumption, which allows the model to experientially learn from iterative activities. To apply FL to UAV data aggregation, edge processing units are often required, either aboard the UAV or via the ground relay. Notably, during operations, information security and communications protocols are vital planning and design considerations. The following sub-subsections delve deeper into some of these.

5.1.1. Onboard deep deterministic policy gradient (DDPG)

Inspired by the trajectory control and device communication protocol dilemmas faced by aerial data aggregators, a DRL-based DDPG scheme

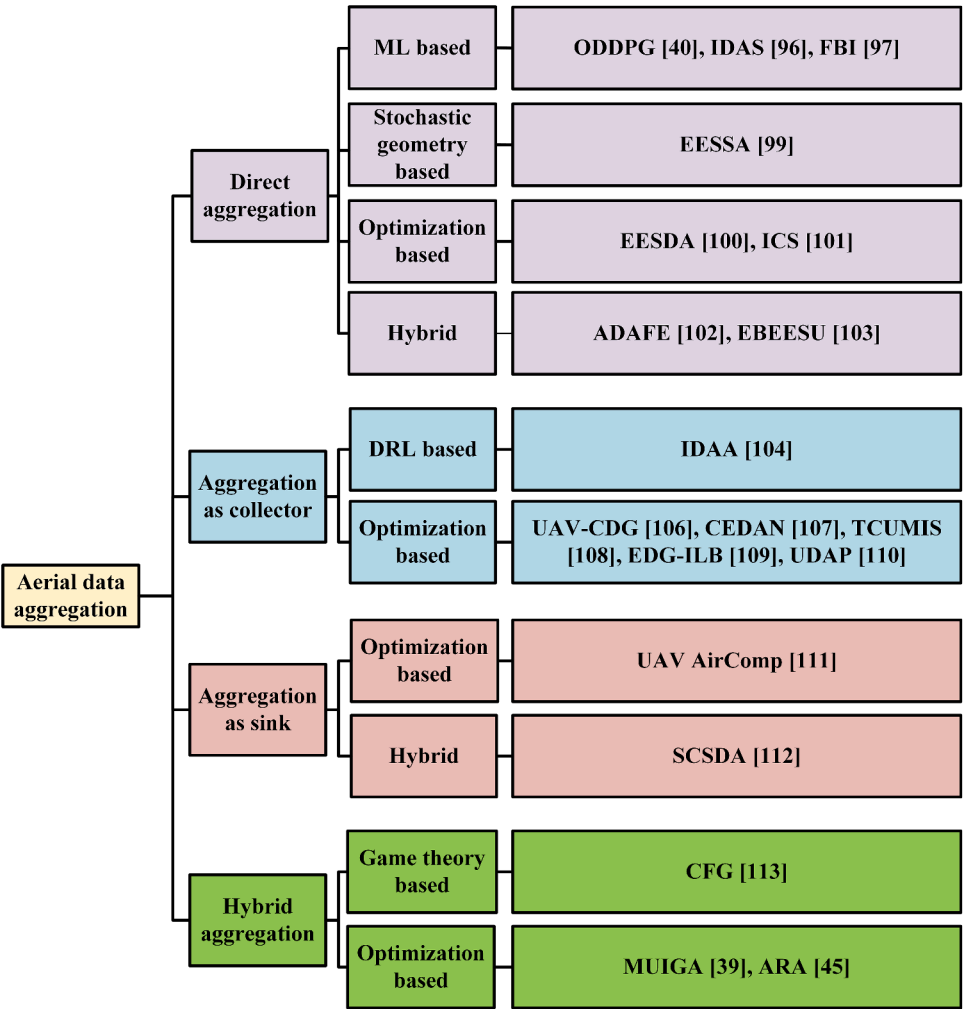


Fig. 2. Taxonomy of aerial data aggregation techniques.



Fig. 3. A typical scenario of UAV-enabled IoT data aggregation where different UAVs perform different roles.

was proposed in [40]. In authentic environmental conditions, the states of IoT devices (e.g., battery level, queue backlog, and channel state) will likely be unknown to the UAV. The DDPG scheme prevents memory overflow and maximizes data aggregation as the UAV cruises among widespread ground IoT devices. In the example of [40], the researchers used Google's TensorFlow to take the joint UAV trajectory and data communication schedule as discrete-time partially observable Markov decision processes, and waypoints and battery levels were formulated as states. The desired action was then to locate the next waypoint and IoT device. The model reward was the aggregated data collected from IoT devices. Because the DDPG model exploits an actor-critic neural network algorithm, mini-batch gradient descent is applied as the loss with respect to UAV speed, packet arrival schedule, and IoT device channel condition for network training and instantaneous trajectory updating.

Lessons learned: In-flight trajectory updating is quite challenging when real-time data are required owing to the inherent fallibility of DRL and FL schemes. However, DDPG provides better packet-loss results than if the free-walk model (the basic random baseline) were used. Moreover, classification errors can be overcome if the model is trained for longer periods. However, DDPG and policy-based gradient methods typically have slow convergence rates; hence, the model performs better when IoT node information is known in advance.

Possible future direction: Based on the application scenario, some of the superior machine learning approaches such as soft actor-critic (SAC) and proximal policy optimization (PPO) can be explored. Combining SAC with DDPG approach can maximize exploration by incorporating entropy into objective function leading to robust policies whereas PPO utilizes multiple updates of data for smooth and stable learning.

5.1.2. Intelligent UAV-based data aggregation strategy (IDAS)

The IDAS scheme was formulated in [96] to restore post-disaster communications among IoT devices using DRL for UAV flight planning and associated tasks. The scheme relies on a tradeoff between data aggregation and energy costs. For training, a state-action reward is sampled from the UAV's previous flight record in mini-batches, and another actor-critic configuration is applied using an update factor to improve learning stability. The example of [96] used four deep neural networks to produce optimized aerial data aggregation schemes.

Lessons learned: The scheme of [96] achieved the highest data aggregation rates under variable UAV speeds among all benchmarks. However, increasing the speed leads to missed nodes and very high energy consumption on behalf of the UAV. Notably, this method is quite suitable for reestablishing communications and aerial data aggregation capabilities after a disaster.

Possible future direction: Rather than using simple system model to describe a post-disaster data aggregation scenario, a detailed model is necessary, which considers device activation and non-overlapping data aggregation for the effective and efficient use of UAV-enabled data aggregation service.

5.1.3. Blockchain-based FL-assisted secure data collection (FBI)

Blockchain-based secure schemes are reliable when dealing with sensitive data. An FL-based blockchain-embedded data accumulation scheme using drones for IoT (FBI) was proposed in [97], in which a two-step data authentication method was applied. The first validation mechanism used a cuckoo filter, and the subsequent mechanism used a Hampel filter to calculate the loss function. Trained models with local IoT data were used to prevent unauthorized access during training. A long short-term memory was used for task preparation, and the UAV maintained NLOS communication most of the time.

Lessons learned: The proposed scheme is successful in providing secure data aggregation from IoT devices as blockchain provides a secure and decentralized network for data transmission and storage scheme. Additionally, the blockchain technique is found to prevent leakage. The only disadvantage is that it did not consider multiple UAV

scenarios and energy consumption requirements during problem formulation and design.

Possible future direction: Blockchain-enabled UAV services envision various possibilities [98]. Because the scheme performs well with a single UAV, multi-UAV scenarios can be exploited in the future. It will be interesting to see whether the same or similar performance can be achieved even in multi-UAV scenarios. Moreover, the scheme should also consider the energy requirements for the mission execution as UAVs are equipped with limited power and have limited flight time.

5.2. Stochastic geometry-based data aggregator

5.2.1. Energy-efficient shared spectrum allocation (EESSA)

The study of arbitrary spatial patterns is known as stochastic geometry and is used to predict communication network performance. UAV-based IoT services are used to predict the coverage and energy consumption of multiple aggregators. Research in [99] demonstrated how the communications spectrum can be time-division multiplexed among ground user equipment (UE), terrestrial BSs, and UAVs when the locations of ground nodes are known. The available frequencies were validated using stochastic geometry. Notably, both UE and IoT device coverages were considered during UAV flight planning for energy management and flight altitude.

Lessons learned: Methods for determining the optimal UAV altitude were developed in terms of lengthening LOS, reducing signal interference, improving energy consumption, and maintaining uninterrupted communications. Moreover, the proposed scheme considers how the existing network infrastructure can benefit from UAV-aided data aggregation. The only drawback was that the study did not consider the impact of the UAV trajectory and mobility in their efficiency equations.

Possible future direction: Similar to finding appropriate altitude of UAV for data aggregation, optimal location selection for hovering may also lead to energy efficiency as shown in some literatures. It is also necessary to determine when IoT nodes are ready to communicate with a UAV.

5.3. Optimization-based data aggregator

Parameter optimization techniques can be classified into two categories: deterministic and stochastic. Deterministic algorithms are based on a set of specific rules for solving problems, whereas stochastic ones follow probabilistic transition rules. Several aerial data aggregation techniques use these optimization techniques and are discussed below.

5.3.1. Energy-efficient spatially correlated data aggregation (EESDA)

Ensuring the longest and most optimized operation times for IoT sensor nodes is challenging. For UAV data aggregation, the research in [100] applied a genetic algorithm (GA), which is one of the popular heuristic approaches, to find optimal solutions under both constrained and unconstrained conditions by imitating the natural selection processes of biological evolution. At each iteration (generation), the GA selects an individual from the population to become a parent of the next round of offspring. This evolutionary process continues until an optimal performing generation is reached. A greedy algorithm comprising two subproblems was applied to manipulate the spatial correlations between energy-efficient UAV flight planning and IoT device scheduling, based on their locations and activation schedules. Because the aggregation points were determined by adopting the nearest neighbor scheme, the TSP problem was solved.

Lessons learned: The proposed scheme demonstrated the best scalability in terms of minimizing energy expenditures and increasing the number of IoT nodes. However, any GA formulation is susceptible to the local minimum problem, in which the generations are forever trapped in a path that diverges from the true optimal solution. This is especially common with greedy algorithms.

Possible future direction: The scheme utilizes the greedy algorithmic approach to solve the aggregation problem and subproblem to solve the optimal hovering location selection for a UAV to aggregate data. Thus, a joint optimization approach with less complexity and execution time can be explored to get a favorable outcome.

5.3.2. Incremental clustering and scheduling (ICS)

ICS [101] offers a collision-free data aggregation scheme based on incremental clustering that coordinates UAV motions and trajectories. Notionally, the CH is offered sufficient time to collect data for aggregation prior to being visited by the UAV. Owing to this aggregation delay, the maximum number of nodes can be harvested. Owing to incremental clustering and scheduling scheme that calculates the transmission schedule of sensors based on the UAV's trajectory and velocity, allowing cluster heads visited later to have more time to collect data. This approach significantly reduces the data aggregation time.

Lessons learned: Realistic UAV-enabled data aggregation scenarios were examined in which device scheduling was well coordinated with the UAV flight path. However, the method did not focus on energy savings.

Possible future direction: Different clustering algorithms can significantly improve data aggregation efficiency. Moreover, the impact of variable UAV velocity can be investigated in aggregation time.

5.4. Hybrid data aggregator

Hybrid techniques obviously increase planning and computational complexity, while also resulting in the best possible solution for the need. Some of these are discussed below.

5.4.1. Aeronautical data aggregation and field estimation (ADAFE)

It is often necessary to minimize the total flight and loitering times of UAVs, and the authors in [102] developed a mathematically mixed-integer NP-hard problem to arrive at a decoupled heuristic solution. For problem formulation, IoT devices were placed across a pre-defined area according to the Poisson point process. The area was then further divided into random subregions. In this scenario, UAVs communicated with IoT using a slotted ALOHA random-access protocol in which activation messages were received by both parties. A multi-cooperative UAV trajectory was designed in which loitering locations were formulated using stochastic geometry. A decoupled heuristic approach was then used to address the NP-hard mixed-integer problem, replacing it with an approximation of the closed-form traveling period.

Lessons learned: The simulation results showed that a clear tradeoff is necessary between hovering and traveling periods to obtain optimal UAV mission times. The travel period approximation resulted in an optimized transmission scheme according to the signal-to-interference-and-noise ratio. However, because the solution requires a lot of time, it is undesirable for real-time problems.

Possible future direction: The work focuses very well on finding optimal hovering point selection for data aggregation. This work can be further extended to find energy association of the UAV with the data aggregation process.

5.4.2. ElectriBio-inspired energy-efficient self-organization model for unmanned aerial ad hoc networks (EBEESUs)

To address the energy consumption problem, a two-level, cluster-based aerial data aggregation algorithm was proposed in [103]. The method combined electrical and bio-inspired models for distributed cluster-based communications, and mobility models were used for multi-UAV scenarios. The cluster-based model considered reduced transmission distances, overhead, and data aggregation schemes, whereas the mobility model comprised loss ratio minimization and the removal of overhead for swarm motions. Simultaneously, an efficient intra-cluster communication technique was devised to minimize the

energy consumption of the two-layer network. Through simulation, the proposed scheme was quite energy efficient and improved the network lifetime. Moreover, the packet-loss ratio and average end-to-end delay were lower than the benchmarks.

Lessons learned: The framework of [103] demonstrated the efficacy of a self-organized, combined electrical and biological data model. The overall scheme extended the network's lifetime by reducing data loss ratios and end-to-end delays while prioritizing energy reduction in the network layer. However, energy consumption in the MAC layer was not considered.

Possible future direction: The proposed model can be tested to see any reduction in aggregation time reduction.

5.5. DRL-based data collector

5.5.1. Intelligent UAV-based data aggregation algorithm (IDAA)

DRL approaches for UAV trajectory and task planning are very common, especially in real-world scenarios lacking prior datasets. Agents in these situations are forced to learn new outcomes based on their interactions with the exposed environment. For example, an intelligent secure data-aggregation scheme for 5G-enabled IoT was proposed in [104], which addressed the placement of IoT sensor nodes in secure and sensitive areas (e.g., military bases and research facilities). In these scenarios, UAVs can act as aerial data aggregators based on the security level of the deployed nodes. In [104], task prioritization and data allocations were assigned based on security requirements, and V-support vector regression (v-SVR) was employed to predict the data collection rate. Next, for optimal route design, a deep Q-learning network (DQN) was used to find the most energy-efficient route to the scheduled nodes based on their security levels. The designated state included the UAV position and the list of remaining tasks. The task-based movement was considered the action, and the reward was the ratio of collected data for aggregation to the energy consumed by the UAV. The simulation results showed that the proposal had a higher sinking ratio and lower energy consumption than benchmarks.

Lessons learned: The 5G network was successfully integrated into secure, UAV-enabled IoT communications. The IDAA model showed superior performance over baselines, in terms of sinking ratio and energy efficiency. However, one drawback of the v-SVR technique is its inability to sustain good performance with large datasets or far-outlying or noisy features.

Possible future direction: Alongside energy efficiency, travel time for the aggregation mission can be studied. Some of the existing authentication mechanism can also be implemented to ensure secure data aggregation [105]. Moreover, a detailed communication model can give a more realistic framework for the proposed method.

5.6. Optimization-based data collector

We have shown that optimized data collection is a foregone assumption in terms of prioritization. However, there are several dedicated optimization techniques that we should cover in this section.

5.6.1. UAV-compressive data gathering (CDG)

Projection-based CDG is used to aggregate data from IoT nodes deployed in hard-to-reach and dense environments. Conceptually, generated data are recoverable as long as the UAV or BS obtains a sparse projection of the originally sensed data. The work in [106] proposed a UAV-aided data aggregation scheme in which a joint problem for compressive data aggregation was modeled with node clustering, forwarding tree construction CH selection, and UAV trajectories. The joint problem had a higher computational complexity than normal, and the execution time increased as the number of nodes per cluster increased. Thus, the problem was decomposed into three subproblems of lower complexity, whose algorithmic efficiencies (e.g., execution time and transmission power) were evaluated in a variety of network scenarios.

Lessons learned: Compressive sensing was found to be a good energy-saving and network lifetime-extending method for aerial data aggregation. Joint problem formulation and decomposition successfully reduced system complexity and computational times. The proposed model gives a clear breakdown of data aggregation using UAV. However, UAV mobility constraints on communication scheduling factors were not considered.

Possible future direction: The proposed method can be implemented with some of the state-of-the-art machine learning algorithms to verify the overall performance improvement. Additionally, multi-agent learning based algorithms can improve the aggregation process more efficiently.

5.6.2. Cost-effective data aggregation for UAV-enabled IoT networks (CEDANs)

By co-designing flight paths and loitering zones, the maximum number of IoT device visits can be achieved. This notion was applied in [107] as CEDAN, which first identified the optimal loitering points, followed by optimizing the TSP flight plan, which was solved using Christofides' approximation. Next, a split function was designed to locate optimized trajectories for the waypoint traversal. Normally, cost functions of this nature are adjusted using the capacitated single-depot vehicle routing problem (CSDVRP). However, because CSDVRP problems are mathematically complex, a heuristic approximation was applied to the UAV problem. The proposed scheme was compared to the Clarke–Wright algorithm and route adjustment without CEDAN.

Lessons learned: CEDAN required fewer UAVs to visit the same number of loitering locations, and IoT device coverage was improved over baselines. However, during problem formulation, UAV scheduling and mobility constraints were not considered.

Possible future direction: The proposed work relies on centralized control of UAV with the BS. This work can be extended for the decentralized execution of UAV commands. Moreover, recent ML approaches can be adopted to see any further improvement in the proposed scheme.

5.6.3. Two-tier communication for UAV-enabled massive IoT systems (TCUMIS)

Multi-tier communication networks for massive IoT systems can utilize UAV capabilities to improve resiliency. This idea was examined by [108], which applied simultaneous trajectory and bandwidth allocations using a two-stage communication model to support many IoT devices. A homogenous Poisson point process distribution of IoT devices was applied, in which data were transferred to designated access points (APs) that assumed the roles of CHs. The combined UAV trajectory and bandwidth allocation applied a successive convex approximation (SCA), a non-convex optimization problem, and an iterative low-complexity algorithm to find sub-optimal solutions. The multichannel ALOHA scheme was used in the first tier, and the time-division multiple-access (TDMA) scheme was used for transmission in the second. The APs established communications with the UAV and transferred their aggregated data for additional processing and ALOHA transmission.

Lessons learned: The formulated optimization problem of [108] increased system throughput, and another iterative suboptimal joint trajectory and bandwidth allocation method was provided for efficiency–performance trade-offs. The use of different schemes increased the overall system throughput and accommodated more aggregated data. Although the method achieved higher throughput than did the benchmarks, UAV trajectory and energy consumption planning were not performed.

Possible future direction: A relative comparison between two-tier communication and direct communication can be carried out to observe any further improvements in aggregation. Moreover, mobility of the ground APs can be introduced to observe the changes in performance.

5.6.4. Energy-efficient data gathering (EEDG)–iterative load balancing (ILB)

A fixed-wing UAV data gathering and aggregation technique was proposed in [109] to emulate the 0–1 multiple knapsack problem (MKPAR) and prioritize the identification of optimal UAV waypoints. Because MKPAR is NP-hard, a two-step maximum energy saving (MES) approximation was used. First, a single knapsack problem was used to maximize ground IoT device energy savings. Then, each UAV's center and radius coordinates were solved using an ILB algorithm, in which trajectories were calculated based on the positions of the ground IoT devices. Additionally, a collision-free communication schedule was designed alongside the method of collecting older data with higher priorities for immediate transmission.

Lessons learned: The scheme saved energy for both the IoT devices and the UAVs, and the data collision avoidance schedule outperformed the two powerful benchmarks. However, the communications schedule was assumed to be ideal, which is unrealistic.

Possible future direction: IoT-UAV communication schedule can be considered to make the proposed problem more realistic.

5.6.5. UAV-based data aggregation protocol (UDAP)

The proposed protocol in [110] introduced a scheme in which a UAV was used as a data mule that visited CHs in a gridded region of interest in which sensors were distributed randomly, and a sink was situated outside the ground-based communication range. The UAV was assumed to have sufficient resources for computation and transmission throughout the region, and all ground nodes were assumed to have identical properties. The CHs were chosen based on node energy consumption and the UAV's travel distance.

Lessons learned: GA-driven optimized trajectory was applied such that the IoT device energy savings were maximized, communications latency was reduced, and the overall network lifetime was increased. However, noting that this paper [100] was one of the earliest attempts at modeling aerial data aggregation, the device communication schedule was not considered in the problem formulation.

Possible future direction: Mission time can be taken into consideration as UAVs are resource-constrained.

5.7. Optimization-based sink

5.7.1. UAV-aided Over-the-Air Computation (AirComp)

The Over-the-Air Computation (AirComp) technique aims to minimize the mean-square error (MSE) of the target function. The method proposed in [111], which was an extended version of the model proposed in [124], employed a UAV as an aerial BS that acted as a sink node when no nearby BS was available. MSE was chosen as the performance metric to reflect the reduction of computation errors based on estimated and target values, where the trajectory of the UAV and the transceiver model were used to formulate the problem. The method addressed the drawbacks of extant sensor power variable studies by reducing the MSE using block-coordinated descent (BCD), which led to the creation of the alternating direction method-of-multipliers (ADMM) scheme, which further addresses UAV trajectory problems.

Lessons learned: Both BCD and ADMM reduce algorithmic and computational complexity, and a reduction in simulation time was achieved in [111]. Notably, this technique works well when the UAV is deployed for long missions. However, the positions of the ground nodes were known in advance to reduce complexity, which is unrealistic.

Possible future direction: multi-UAV deployment scenarios can be studied for the extension of the proposed scheme. Moreover, IoT-UAV communication schedule can also be studied to make the formulated problem more realistic.

5.8. Hybrid sink

5.8.1. Sparsity-optimized and compressed sensing-based spatiotemporal data aggregation (SCSDA)

As noted, to perform hybrid roles, UAVs must be well-equipped with sufficient power and resources. Moreover, ensuring data security and privacy is a common hybrid challenge. The SCSDA scheme was devised in [112], in which a UAV-enabled framework using a disaster-semantic blockchain (DSB) data collection scheme was formed by integrating sensor, aggregated, and semantic disaster-related data. In normal cases, the UAV communicates with the CH using Wi-Fi, and ZigBee is used to communicate with isolated nodes. A spatiotemporal sparse dictionary matrix using data from multiple UAVs was leveraged to reconstruct originally sensed data and improve the reconstruction process. To verify the identity of the designated UAV, a Merkle tree-based authentication mechanism and a popular blockchain model were applied.

Lessons learned: smaller amounts of data were needed for processing and transmission, which reduced energy consumption and bandwidth requirements. The method resulted in longer IoT lifetimes and provided near-optimal reconstructed data with negligible errors. The Merkle tree-based secure UAV recognition method ensures that DSB data are distortion-free, and only registered UAVs can collect data from the sensors. The accuracy of the reconstructed data was statistically verified, and real-time data sensing and processing were demonstrated. However, the UAV mobility model was not examined.

Possible future direction: Although the proposed work considers UAV as a static sink node for data aggregation, the UAV mobility can be taken into consideration for further extension of the proposed work.

5.9. Game theory-based hybrid role

5.9.1. Coalition formation gaming (CFG)

Game theory is often used to study scenarios in which parties (players) make interdependent decisions, in which the decision of one player is based on that of another. Players can have similar, opposing, or mixed interests and their own reward functions. For multi-UAV scenarios, resource-sharing for aerial data aggregation was the winning objective in [113], which applied coalition formation gaming (CFG) to the behavioral patterns of cooperating UAVs to achieve the target task. Energy efficiency was achieved using store-carry-forward (SCF) or hop-by-hop transmission schemes, resulting in four unique strategies for intra-UAV communications. Among these, optimal UAV coalition formations were built in preference, Pareto, utilitarian, and merge-rule ordering modes.

Lessons learned: The method demonstrated bona fide cooperative communications among UAVs and reacted well to UAVs joining or leaving the topology. Moreover, SCF performed better than SCG in terms of energy savings, and based on two possible scenarios, utilitarian ordering was superior to Pareto in terms of converging to a coalition structure. However, the study did not include trajectory and energy efficiency in its multi-UAV paradigm.

Possible future direction: The proposed scheme considers data aggregation in air-to-air scenarios. The inclusion of ground IoT nodes into the existing work can be an extension of the proposed scheme.

5.10. Optimization-based hybrid role

This subsection addresses optimization with hybrid UAV and IoT device configurations.

5.10.1. Multi-UAV information gathering application (MUGA)

The authors in [39] tackled the data aggregation problem of area mapping and target-tracking scenarios using a mixed-integer nonlinear programming (MINLP) method to determine the optimal energy expenditure for communication and computation in multi-UAV scenarios. UAVs were assigned as sensors, relays, or aggregators using

self-organizing, multi-hop, or hierarchical clustering.

Lessons learned: Simulation results showed energy savings of ~40% and ~60% in energy consumption and target tracking, respectively. The proposed scheme performed better when the area mapping scenario was considered for data aggregation. The key takeaway is that the energy consumption of data aggregation can be mitigated with properly tuned data aggregation and clustering schemes. The disadvantages of this method can be stated as communication schedule between the UAV and BS was not studied, and algorithmic efficiency was not examined for centralized and distributed UAV performance.

Possible future direction: Communication schedule among UAV, sensors, and BS can be taken into consideration to make the proposed work more vibrant one.

5.10.2. Aggregators-aided random access (ARA)

IoT have drawn considerable attention as building blocks for smart cities. Incorporating a space-air-ground integrated network with vehicular user equipment (VUE) can be applied to aerial data aggregation tasks to avoid data collisions and improve overall communication performance. The ARA model was proposed in [45] to account for UAV-to-IoV communications using slotted ALOHA, which is an improvement over standard ALOHA in terms of reducing data collisions and information loss. The scenario in which the UAV acts as a data aggregator and transmits the data to a BS was investigated using a semi-Markov chain. To determine the model's efficiency, the novel average data-to-overhead ratio (ADOR) was calculated based on UAV altitude, data aggregation performance, and optimized aggregation period.

Lessons learned: Both simulation and empirical examinations showed that the altitude and aggregation period influenced aerial data aggregation the most. As both the VUE and BS distributions imitate the Poisson distribution, they are also applicable to realistic scenarios. However, the work of [45] did not associate energy consumption with process execution, and energy-saving mechanisms were not considered. Moreover, the algorithmic complexity of the proposed scheme was not examined.

Possible future direction: A more realistic vehicle mobility model can be incorporated to understand the data aggregation scenario rather than relying on statistical distribution of task arrival.

6. Comparison of data aggregation techniques

This section discusses the performance comparisons among the various aerial data aggregation techniques from Section 5. For the techniques, we first summarize the models and their application scenarios herein, alongside their respective advantages and limitations as shown in Table 4.

In [40], the authors considered crop growth monitoring in farmlands where ground IoT sensors were distributed. A UAV acted as a data aggregator, and its combined flight path and data collection schedule were built using a DDPG model. Although the proposed scheme performed better than benchmarks in terms of packet loss, the model convergence rate took longer than the other DRL methods. Moreover, extra node information is required in advance, which is not always realistic.

Another DRL approach was taken in [96] in which the UAV provided support to an aggregator backbone in a post-disaster scenario. To maximize energy savings, an efficient UAV route was designed by trading costs with data aggregation ratios. This scheme showed better energy savings than benchmarks, but higher performance could be achieved if a longer training period was provided. The work in [97] demonstrated secure data aggregation in a remote region using FL and blockchain. This model has made good advancements toward secure aerial data aggregation and may be further improved by incorporating multi-UAV scenarios and focused energy savings.

The authors in [99] illustrated the importance of infrastructure and farmland water monitoring using UAVs as aggregators while applying a stochastic geometric approach. The formulated problem focused on a shared UAV transmission strategy alongside a preexisting telecommunications infrastructure. The energy efficiency for aerial data aggregation was clearly demonstrated. However, trajectory, IoT device activation, and mobility model of UAVs were overlooked.

In terms of device activation, the study of [100] used a GA and a greedy algorithm to solve the UAV itinerary problem. The proposal aimed to provide energy efficiency for aerial data aggregation. However, the downside of the GA approach is that it often fails to converge to the optimal solution.

The study of [101] focused on the data aggregation of IoT sensors deployed over a wide region using an incremental clustering scheme for a collision-free transmission schedule and flight time minimization. Although the model provided a shorter aggregation period, energy-related studies were overlooked. The authors of [102] studied loitering and mission times in dense IoT environments for sensing purposes, aiming to determine minimal flight and visiting times. However, it failed to address energy-related issues for UAVs as well as for communication purposes.

The algorithm proposed in [103] focused on a generic monitoring application in which the mobility model was built on a combination of electrical and biological methods. A cluster-based UAV communication model with a two-stage energy-saving data aggregation scheme successfully reduced the overall energy consumption and increased the overall network lifetime, but energy consumption and MAC-layer leakages were not considered.

Security is an obvious concern, as addressed in [104]. The problem formulation was based on the IoT security tier, in which a DQN technique was applied for path planning. The resultant IDAA scheme demonstrated better performance than the baselines in terms of sinking ratio and energy efficiency, but the v-SVR model was unsuitable for large datasets. Moreover, there is a risk of underperformance when the data points exceed the training dataset and noise contamination exists.

The scheme in [106] focused on UAV-aided data aggregation in rural and urban areas, using a heuristic approach for compressive sensing. A joint cluster model based on CH selection and UAV trajectory planning was devised. To lower complexity, the problem was broken down into subproblems, and a heuristic approach was used to solve them. The main advantage is that the method avoided long-range and multi-hop communications, and the execution time was significantly reduced. The disadvantage is that it did not consider UAV scheduling and mobility constraints.

Another recent approach [107] focused on minimizing the total travel time while maximizing IoT device coverage for geographical monitoring and sensing. To reach the target objective, a CSDVRP method for solving the TSP was used, which performed better in terms of device coverage and performance. However, energy efficiency was not considered.

In [108], the authors formulated a two-tier data aggregation scheme with simultaneous trajectory and bandwidth allocations. The formulated problem was solved using SCA, which increases the average system throughput over contemporary schemes. The drawback is that the UAV trajectory and energy-saving mechanisms were not examined.

The approach in [109] targeted a generic monitoring area in which IoT devices were randomly distributed along the ground. The device association problem was addressed by MES, and the ILB approach was used to determine the optimal fixed-wing UAV radius for loitering. The proposed work emphasized collision-free data, energy savings, and data freshness, and outperformed two contemporary algorithms. However, the model also assumed an ideal communication schedule, which is unrealistic.

The model of [110] used a GA to identify the shortest path for the UAV aggregation of data transmitted from ground CHs. Distributed nodes were deployed for intruder detection in the area of interest, and an

optimized traversal distance was formulated for the UAV, which was validated by extant benchmarks. However, it should be noted that heuristic algorithms like GA do not ensure convergence all the time. Moreover, device communication scheduling was not considered.

The authors of [111] focused on data aggregation in remote and rough areas for which communication infrastructures were unavailable. The scheme was an AirComp-based MSE minimization technique that provided reduced mission and simulation times. It was limited in that it did not consider multi-UAVs or mobile sensor prediction errors.

The work in [112] illustrated a post-disaster scenario that relied on semantic knowledge. Sensor data were reconstructed using CS, and UAV authentication was performed using a Merkle tree. The study provided blockchain-enabled compressed data aggregation, in which only registered UAVs were allowed to collect, reconstruct, and distribute data. This model provides a pathway to secure, distortion-free, and real-time data communications. However, it did not examine UAV trajectory and mobility mechanisms.

In [113], small- and large-scale UAV operations were discussed in terms of SCS and SCF strategies based on coalition game theory, which sought the most energy-efficient multi-UAV aerial data aggregation. The scheme used valid strategies, and UAV swarm formation was clearly effective. However, multi-UAV trajectories and energy-efficiency features were overlooked.

Target tracking and areal mapping applications using UAV were studied in [39] which proposed energy-saving aerial data aggregation for two scenarios and examined the impact of various parameters on MINLP formulation. The study showed that energy consumption can be lowered by implementing proper data aggregation and clustering algorithms. The downside is that the communication schedule between the UAV and the BS was not examined. Additionally, the algorithmic efficiency of UAV task performance was not tested in centralized or distributed scenarios.

Roadside vehicular data using UAV technologies were studied in [45], which presented a novel aerial aggregation performance metric that is also useful for mobile VUEs. However, the study did not examine the energy consumption of the UAV or its aerial data aggregation mechanism.

Table 5 lists these simulation parameters and their algorithmic approaches. Parameters include UAV type, simulation environment area, altitude, velocity, number of ground units, whether mobility was considered, communication bandwidth, carrier frequency, and simulation tools. Table 6 provides a comparison of the models reviewed to illustrate the different achievements based on their respective objectives. Only the most common parameters were compared.

The ODDPG [40] scheme aimed to formulate an optimal cruise control mechanism to provide a backbone for aerial data aggregation based on the communication schedule and UAV battery level. Performance was demonstrated in terms of packet loss rate.

The IDAS [96] scheme measured the energy efficiency of the UAV trajectory using an aggregation ratio. The FBI [97] scheme improves the security of aerial data aggregation with blockchain and UAV identification mechanisms. The efficacy of the method was demonstrated in terms of the data transmission rate.

In EESSA [99], the authors examined energy and spectral efficiencies. They showed different coverage probabilities as evaluation parameters and determined how to increase the network lifetime. EESDA [100] was used to formulate an energy-efficient trajectory for data aggregation, considering an optimized communication schedule. The evaluation parameters were presented in terms of energy efficiency. The ICS [101] scheme provided a data collection communication schedule and clustering scheme for ground nodes, and UAV trajectories were built to fit the schedule. The model upheld its desired performance thresholds in terms of reducing end-to-end delays.

ADAFE [102] aimed to determine the minimal mission time for aerial data aggregation. Hence, a proper trajectory plan was examined to ensure the minimal MSE based on the loitering radius and flight

Table 4
Comparison of aggregation techniques in terms of key features.

Scheme	Application scenario	Algorithmic Approach	Main focus	Advantage	Limitation
ODDPG [40]	Farmland for crop growth monitoring	DRL (DDPG)	A joint cruise and aggregated data collection schedule is devised using deep deterministic gradient policy.	The proposed model performs better than benchmark in terms of packet loss rate after 50 episodes. Moreover, classification errors can be overcome if model is trained longer.	DDPG or policy based gradient methods typically have slow convergence rate and the model works best when the information of IoT nodes is known.
IDAS [96]	Restoring post-disaster communication	DRL	Energy-saving aerial data aggregation scheme achieved by trading off between cost and data aggregation ratio.	The scheme shows energy-efficiency when compared with other benchmarks.	Requires longer training time for better performance.
FBI [97]	Remote data accumulation region	FL, Blockchain	A secure FL based data aggregation technique using drone is illustrated and aggregated data is stored in blockchain.	The scheme prevents privacy leakage and reduces connectivity problems.	Impact of swarm of drone was not studied. Moreover, energy related issues were ignored.
EESSA [99]	Infrastructure monitoring, utility metering, farmland water management.	Stochastic geometry	Devises an energy-efficient time-division duplexing (TDD) shared transmission scheme with optimal altitude for aggregation operation alongside existing telecom operation.	Energy-efficiency and shared transmission mode for aerial data aggregation is achieved.	Study on UAV trajectory, mobility mechanism and IoT device activation was not emphasized.
EESDA [100]	Generic area of interest	Genetic algorithm, greedy algorithm	An energy-efficient aerial data aggregation scheme for IoT devices with device activation model and appropriate path planning is proposed.	Energy-efficiency is achieved compared to other benchmarks.	Genetic algorithm often fails to find optimal solution and computationally expensive.
ICS [101]	Sensor network deployed in wide region	Incremental algorithm	UAV-enabled data aggregation with incremental clustering scheme is proposed.	Reduces aggregation period of UAV as visitation time is minimized.	Energy studies are not included in problem formulation.
ADAFE [102]	Dense IoT network for environmental sensing	Stochastic geometry, graph theory, MINLP	Develop a minimized hover and travel time for ground IoT device for aerial data aggregation by UAV.	Established the relation of hovering and travel duration with overall mission duration.	The proposed algorithm takes longer to solve which is undesirable for real-time scenarios.
EBEESU [103]	Generic monitoring application	Electro-Bio inspired	A combination of Electrical-Bio inspired mobility and cluster-based UAV communication model with two-stage energy saving data aggregation scheme.	Reduced network energy consumption with increase network lifetime. Packet loss ratio and end-to-end delay mitigated.	Energy consumption and leakages in MAC layer communication were not considered.
IDAA [104]	Confidential or secured area	V-support vector regression, DRL	Based on the security tier of IoT devices, aerial data aggregation is performed, using DQN for path planning of UAV.	IDAA demonstrates better performance than the baseline algorithms in terms of sinking ratio and energy efficiency.	v-SVR is not suitable when the data set is large. Moreover, there is a risk of underperformance if the data points exceed the training dataset and contaminated by noise.
UAV-CDG [106]	Rural and urban area with fewer buildings	Heuristic, Compressive Sensing	Formulate joint cluster formation, CH selection and UAV trajectory planning for CDG. In order to lower the complexity, the problem is broken down into sub-problems.	Avoids long range and multi hop communication. Execution time is significantly lowered.	Scheduling and mobility constrain of UAV not considered.
CEDAN [107]	Geographical area monitoring (e.g., smart waste and street light management)	CSDVRP, Non-convex	The scheme comprises minimization of total travel time while maximizing the covered IoT device. To reach the target objective, a CSDVRP for Travelling Salesman Problem (TSP) has been formulated.	The scheme works better in terms of covering IoT devices, outperforms contemporary methods	Concept of energy efficiency was not considered in this work.
TCUMIS [108]	Ground IoT sensors	SCA	Formulates a two-tier data aggregation scheme with simultaneous trajectory and bandwidth allocation.	The scheme increases the average system throughput in comparison to contemporary schemes.	UAV trajectory and energy saving mechanisms are not studied
EEDG-ILB [109]	Generic monitoring	MES, ILB	A data collision-free and energy saving UAV-IoT communication scheduling with data freshness algorithm is proposed for fixed-wing UAV network.	Proposed method outperforms other two contemporary algorithms in terms of energy saving of ground IoT devices.	Communication schedule between UAV and ground IoT devices is assumed ideal which is not realistic.
UDAP [110]	Military application for intruder detection	Genetic Algorithm	Energy-efficient UAV trajectory is designed for aerial data aggregation using genetic algorithm (GA).	Optimized traverse distance for UAV is formulated which is validated by other benchmarks.	Device communication schedule was not taken into account in problem formulation.
UAV AirComp [111]	Wild area monitoring	BCD-ADMM	AirComp system based aerial data aggregation scheme to minimize time average MSE is proposed.	Proposed scheme minimizes the MSE with reduced mission time and simulation time.	Multi-UAV aerial data aggregation scheme and prediction error of mobile sensors were not considered.

(continued on next page)

Table 4 (continued)

Scheme	Application scenario	Algorithmic Approach	Main focus	Advantage	Limitation
SCSDA [112]	Remote monitoring for natural disaster	Compress sensing, Merkle tree, linear programming	Blockchain-enabled UAV-assisted compress data aggregation technique is studied where only registered UAV in the system allowed for data collection and reconstruction.	Distortion-free real-time data communication and data node transfers data only to registered UAV.	UAV trajectory or mobility mechanisms were not studied.
CFG [113]	Small and large-scale UAV operation	Game Theory	Presents SCS and CFS strategy based on coalition game theory for energy-efficient multi-UAV aerial data aggregation.	Validation of each strategy based on operation scale and UAV swarm formation clearly explained	UAV trajectory and energy-efficiency from multi-UAV to multi destination studies are overlooked.
MUIGA [39]	Target tracking and area mapping	Mixed Integer Non-Linear Programming	Proposes energy-saving aerial data aggregation scheme on two different scenario and impact of various parameters on MINLP formulation.	Energy consumption can be lowered by implementing proper data aggregation and clustering algorithms.	Communication schedule between UAV to BS is not studied. Additionally, algorithmic efficiency is not examined when UAV performs the task in centralized or distributed manner.
ARA [45]	Roadside vehicular data	Semi-Markov Chain	Average aggregated packet number and efficiency metric of aggregation technique is presented. Additionally, optimal aggregation period and UAV altitude for aggregation evaluated.	The proposed scheme can be considered for mobile VUEs.	Energy association with the scheme was not explained.

distance. The EBEEUS scheme [103] focused on energy efficiency based on an optimal number of UAVs, and model performance was measured in terms of network lifetime and packet loss.

The IDAA [104] scheme sought to provide security for aerial data aggregation and explored energy-efficient trajectories for data aggregation. Model performance was validated in terms of the sinking ratio.

The creators of UAV-CDG [106] formulated optimal UAV trajectories for data aggregation while reducing computational times by decomposing the original complex problems into subproblems while applying an appropriate node clustering technique. Model performance was validated by reducing the total transmission power of the IoT nodes.

Using CEDAN [107], the authors aimed to minimize mission times while maximizing device coverage. They devised a communication schedule in between UAV and IoT nodes along with an optimal UAV trajectory that improved overall energy savings. With TCUMIS [108], the authors developed a combined trajectory and bandwidth allocation approach for two-tier data aggregation using UAVs. The performance of the proposed scheme was measured using the average system throughput.

EEDG-ILB [109] focused on UAV trajectory optimization and device coverage maximization while considering the communication schedule and node activations. The results showed improved energy efficiency. The authors of UDAP [110] aimed to achieve energy efficiency by optimizing UAV trajectories. The results were validated by comparing them with end-to-end delay benchmarks. The UAV-AirComp concept [111] aimed to minimize the MSE of data aggregation by optimizing the UAV trajectory. The model performed better than baselines in terms of energy savings and mission time. The SCSDA scheme [112] focused on the security of aerial data aggregation, showing promise in terms of increased overall network lifetime and optimal deployment.

The CFG model [113] focused on energy efficiency in multi-UAV scenarios. The findings demonstrated that a UAV should not join or leave the cluster to perform cooperative aggregation. The MUIGA scheme [39] emphasized energy efficiency and computational complexity reduction, and ARA [45] introduced a new metric for measuring the efficacy of aerial data aggregation. ARA also predicted the optimal altitude from maximum aggregation.

After studying the existing literature and exploring the recent works in UAV-aided data aggregation scenarios, it can be observed that the key idea is to find optimal hovering location or loitering point which can serve the maximum number of IoT devices and gather data at a time. To achieve that feat, careful trajectory planning and energy saving waypoint should be designed to utilize the available resource at disposal for the UAV while ensuring the minimal flight time. Most of the existing works explore the option of single UAV-based data aggregation scenario. As explained earlier, the introduction of the multiple IoT nodes in the environment ultimately leads to a larger number of waypoints for the UAV to visit, which increases the complexity of the model further. In some cases, some IoT nodes need to send data earlier than the others due to the overflow of buffer memory or emergency data such as event-driven fire detectors. To address this issue, multiple UAVs are deployed to cover the vast area. However, cooperation among the UAVs is necessary so that multiple waypoints are not visited by the same UAV, which will ultimately lead to the wastage of resources. Therefore, adaptive cooperation amongst UAVs is necessary to support such actions [114]. Multi-agent DRL techniques have shown to tackle such scenarios [118]. Moreover, the deployment of the optimal number of UAVs is also necessary not to further increase the complexity of the proposed algorithm.

While studying the existing scheme, we came across two important aspects of aerial data aggregation: reliability and scalability [107]. Reliability is an important system performance parameter that refers to the competency or guaranteed performance level of any system under given constraints. The reliability and scalability of aerial data aggregation entails the number of ground IoT devices covered by UAVs at a given time. Alternatively, scalability provides an idea of how well the

Table 5

Comparison of aggregation techniques in terms of parameters used in the performance study.

Scheme	UAV Type	Simulation Area	Altitude	Velocity	Number of UAV	Number of Users	Mobility	Communication BW	Simulation Tool
ODDPG [40]	Rotary	1000 × 1000 m ²	–	–	Single	[100,200]	–	–	Google Tensorflow
IDAS [96]	–	1000 × 1000 m ²	–	[20, 90] km/h	[4,11]	[100,240]	–	–	Python
FBI [97]	–	–	–	–	Single	10	–	–	Pytorch, PySyft, Opacus, Python
EESSA [99]	Rotary	–	[50,100,150,200,250] m	–	Multiple	–	–	2 GHz	Monte Carlo Simulation, 3GPP toolkit
EESDA [100]	–	1 × 1 km ²	–	15 m/s	Single	100	–	2 GHz	–
ICS [101]	–	500 × 700m ² 700 × 700m ²	–	40 m/s	Single	500–900	–	–	–
ADAFE [102]	Rotary	100 × 100m ²	20 m	20 km/h	Multiple	1000	Yes	200KHz	–
EBEESU [103]	–	1000 m radius	–	–	Multiple (Swarm)	60	Yes	2.4 GHz (IEEE 802.11)	BulkSendApplication
IDAA [104]	–	3000 × 3000 m ²	10, 50 m	1, 8 m/s	[5, 12]	250–500	–	2.45 GHz	Python
UAV-CDG [106]	–	900 × 700 m ²	50 m	–	Single	20,30,50 40–100	No	–	CPLEX and JAVA-based simulator
CEDAN [107]	–	1.5 × 1.5km ²	–	–	3–8	500	Yes	2 GHz	–
TCUMIS [108]	–	350 m radius	100 m	30 m/s	Single	–	–	1 MHz	CVX (MATLAB)
EEDG-ILB [109]	Fixed	600 × 600 m ²	100–300m	–	Multiple	300, 500	–	2 GHz	–
UDAP [110]	–	250 × 250 m ²	–	20 m/s	Single	[150, 260 410, 590]	–	–	OMNeT++
UAV-AirComp [111]	Rotary	400 × 400m ²	100 m	20 m/s	Single	50	Yes	–	CVX Solver
SCSDA [112]	Rotary	1400 × 1200 m ²	–	–	1–8	120	–	2.4 GHz	–
CFG [113]	–	2000 × 2000 m ²	–	–	500	10–30	–	1 MHz	MATLAB
MUIGA [39]	–	3000 × 3000 m ²	–	10–30 m/s	Multiple	5	Yes	–	MATLAB, OPTI TOOLBOX, SCIP
ARA [45]	Rotary	800 × 800 m ²	10–200	–	Multiple	0.02 (VUE density)	–	1 MHz	MATLAB

Note: “–” means that the information is not mentioned in the corresponding literature.

scheme performs when variables change. That is, it is the ability to maintain network performance when limiting factors begin to build up. A well-designed scheme might work well for a single UAV scenario while failing in multi-UAV applications.

7. Open issues and research challenges

In addition to the state-of-the-art design considerations, open issues and research challenges remain to be addressed to achieve seamless and loss-free data aggregation. We discuss them in this section.

7.1. Aggregation schedule and waypoint design based on battery level

It is well known that UAVs and IoT sensor devices are energy constrained and that UAV flight times are relatively short. Thus, it is necessary to streamline and optimize IoT device activations and UAV waypoint planning based on battery life. Furthermore, as one UAV is returning to its central charging station, another could be deployed to continue the aggregation task seamlessly. Additionally, for time-constrained IoT devices with limited resources, the trajectory planning of UAVs should be carefully designed [116].

Event-driven IoT devices (e.g., fire detectors) are subjected to random activation and are expected to expend more energy than other devices. The authors in [115] discuss the potentiality of aggregation

schedule and waypoint design based on multi-agent DRL, where drones share channel state and network information with their neighboring UAVs. Incorporating long-short term memory (LSTM) and DDPG together can increase learning accuracy as well as resource sharing at a faster rate at the expense of increasing overhead. Moreover, introducing the LSTM module is effective at capturing sequential data as it can retain past observation due to the introduction of memory cells (long-term dependencies), leading to more accurate prediction. The model is trained offline and has further scalable option for online learning. Therefore, aggregation scheduling and waypoint design must adopt commensurate reaction capabilities.

7.2. Age-of-information (AoI) incorporation

The recent AoI construct has captured considerable attention as it seeks to ascertain the freshness of data packets [117] based on the delivery time of the previous packet [118]. Several studies have examined AoI-related communication, trajectory planning, and data-acquisition schemes for UAVs [119–121]. Notably, the AoI is treated as a metric that must be reduced as much as possible based on packet delivery while simultaneously minimizing energy expenditures [58]. Notably, security risks increase with longer AoI.

Table 6

Comparison of aggregation techniques in terms of objective parameters.

Scheme	Objective Parameter																			
	Trajectory/Path Planning	Energy-Efficiency	Spectral Efficiency	Altitude	Mission time	Communication schedule	Aggregation/Sinking ratio	Data transmission rate	Security	Optimal No of UAV	Mobility Model	Network lifetime	Packet Loss	Bandwidth allocation	End-to-end delay	Throughput	Computational time	Mean Square Error	Node Clustering	UAV device coverage
ODDPG [40]	H	—	—	—	—	M	—	—	—	—	—	—	P	—	—	—	—	—	—	—
IDAS [96]	M	H	—	—	—	—	P	—	—	—	—	—	—	—	—	—	—	—	—	—
FBI [97]	—	—	—	—	—	—	—	P	H	—	—	—	—	—	—	—	P	—	—	—
EESSA [99]	—	H	P	P	—	—	—	—	—	—	—	M	—	—	—	—	—	—	—	P
EESDA [100]	H	P	—	—	—	M	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ICS [101]	M	—	—	—	—	H	—	—	—	—	—	—	—	—	P	—	—	—	M	—
ADAFE [102]	M	—	—	—	H	P	—	—	—	—	—	—	—	—	—	—	—	M	—	—
EBEESU [103]	—	H	—	—	—	—	—	—	—	M	M	P	P	—	P	—	—	—	—	—
IDAA [104]	M	M	—	—	—	—	P	—	H	—	—	—	—	—	—	—	—	—	—	—
UAV-CDG [106]	H	P	—	—	—	—	—	—	—	—	—	—	—	—	—	—	M	—	M	—
CEDAN [107]	M	P	—	—	H	M	—	—	—	—	—	—	—	—	—	—	—	—	—	H
TCUMIS [108]	H	—	—	—	—	—	—	—	—	—	—	—	—	H	—	L	—	—	—	—
EEDG-ILB [109]	H	P	—	—	—	M	—	—	—	—	—	—	—	—	—	—	—	—	—	H
UDAP [110]	M	H	—	—	—	—	—	—	—	—	—	—	—	—	—	L	—	—	—	—
UAV-AirComp [111]	M	P	—	—	P	—	—	—	—	—	—	—	—	—	—	—	P	H	—	—
SCSDA [112]	—	—	—	—	—	—	—	—	H	P	—	P	—	—	—	—	—	—	—	—
CFG [113]	—	H	—	—	—	—	—	—	—	P	—	—	—	—	—	—	—	—	—	—
MUIGA [39]	—	H	P	—	—	—	—	—	—	—	—	—	—	—	—	—	P	—	—	—
ARA [45]	—	—	—	P	—	—	H	—	—	—	—	—	—	—	—	—	—	—	—	—

Note: H: Main objective in problem formulation; M: Joint/Sub-objectives studied at a moderate level; P: Performance evaluation parameter. “—” means that the information is not mentioned in the corresponding literature.

Note: H: Main objective in problem formulation; M: Joint/Sub-objectives studied at a moderate level; P: Performance evaluation parameter. “—” means that the information is not mentioned in the corresponding literature.

7.3. AirComp adoption

The principal of AirComp is to exploit the waveform superposition property in the MAC layer [122]. This differs greatly from conventional multiple access schemes in that AirComp enables all devices to access all radio resources simultaneously without orthogonality [123]. Because the scheme uses analog modulation, it suffers from signal distortions. However, many of these issues can be addressed by reducing the Gaussian MAC MSE distortion. As aerial data aggregation research is still in its early stages [111,124], AirComp technologies can be more expediently incorporated to develop faster and more accurate data aggregation schemes.

7.4. Digital twin-assisted aerial data aggregation

The introduction of digital twin (DT) techniques to aerial data aggregation will be groundbreaking in terms of troubleshooting and problem-solving [125]. The concept uses simulated systems in a simulated environment so that optimization strategies can be tested to determine how they might affect overall performance. Moreover, emerging technologies (e.g., blockchain, fog and edge computing, virtual modeling, simulation techniques, and ML) can be incorporated in unique ways without tremendous expenditures [126]. Provided that the DT provides a high-fidelity replication of a real-world scenario, historical and environmental data can be applied to virtual IoT for DRL model applications in support of real-time decision making [70]. Moreover, for large-scale multi-UAV scenarios, aircraft collision avoidance is a significant challenge, as is finding the appropriate IoT nodes [127].

Implanting DT can help achieve real-time data aggregation and aid with the predictive analysis to mitigate communication issues. Additionally, the DT-aided framework with centralized training and decentralized execution can cope with the multi-UAV scenario in a cooperative manner, which aims to speed up the multi-agent DRL training process [128]. This ensures faster convergence and better decision making for the multi-agent DRL scenario.

7.5. Incorporation of transfer learning

Because UAVs often suffer from low processing power and battery constraints, they cannot readily handle the training workload of high-complexity ML models. Because transmitting raw data can lead to data breaches and privacy concerns, transfer learning (TL) has demonstrated its efficacy in alleviating UAV-enabled communication shortfalls [129]. Unlike traditional ML techniques, TL does not require a training dataset; rather, it relies on an existing labeled or trained model that is similar to the proposed model. The process greatly decreases training time, and the only problem is finding a suitable model that fits the target environment. The knowledge that needs to be transferred in transfer learning-aided aerial data aggregation depends on the specific task and domain being considered. However, in general, the goal is to transfer knowledge that is useful for improving the performance of a machine learning model in the target domain [130]. Task-specific knowledge such as the location of the IoT nodes for data aggregation can be learned in the source domain and transferred to the target domain. This knowledge can be used to improve the performance of a machine learning model for surveillance applications. Hence, building a model suitable for aerial data aggregation is an important task for future work.

7.6. Joint aerial data aggregation and task offloading

At the network edge, where images and videos are most often generated, heavy resources are often applied to accelerate data transfers and ensure QoS [131]. In [108], the authors proposed a data offloading method in support of emergency disaster recovery so that instantaneous decisions could be facilitated. Additionally, ML-based multi-access edge computing can help reduce latency and energy consumption [132]. Thus, designing appropriate task-offloading schemes for aerial data aggregation is needed, especially when lives are at stake.

7.7. Edge intelligence

Edge intelligence derives from the concept of combining edge computing and ML so that the UAV acts as a mobile edge server containing extra onboard processing power [133]. This is particularly useful for task offloading and latency reduction. However, carrying and employing extra processing units is very burdensome for UAVs. Thus, the combination of edge intelligence and aerial data aggregation requires new pathways to meet the appropriate trade-offs needed for edge intelligence support.

7.8. Blockchain-assisted aerial data aggregation

Secure and privacy-aware data aggregation schemes can be ensured by employing blockchain technologies [134] for secure data aggregation. As proposed in [97], UAV-enabled IoT data aggregation applies blockchain data that can be verified prior to processing. Such methods have shown high potential to ensure privacy-aware data aggregation, particularly for military operations and smart grid management systems.

7.9. Drone-based autonomous aggregation

Autonomous UAV operations are desirable and have been long examined, such as with vision-based UAV navigation and surveillance [135]. UAVs rely on onboard processing power for visualization and

mapping, and sudden changes in weather can disrupt operations. Hence, autonomous systems can ensure seamless reactions and data aggregation continuance [136]. Autonomous systems are also applicable to obstacle avoidance [137], especially when there is no prior knowledge of the surrounding environment [138].

8. Conclusion

In this article, UAV-aided data aggregation schemes for IoT networks were extensively surveyed, and several challenging issues were comparatively discussed. A variety of configurations and implementation methods were summarized, and the design issues were analyzed. A taxonomy of data aggregation techniques was also established. The existing aerial data aggregation techniques were qualitatively compared in terms of key operational features, advantages, limitations, performance parameters, and objective parameters. We expect that researchers can build upon this compilation to address the remaining open issues and research challenges while driving towards sustainable aerial data aggregation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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